Exploration of Gene Co-Expression and Functional Modules in Pseudomonas aeruginosa PA14

George David Apostolidis[1*] and Michail Tsiledakis[1*]

[1*]Wageningen University & Research, Droevendaalsesteeg 4, Wageningen, 6708 PB, Gelderland, The Netherlands.

*Corresponding author(s). E-mail(s): george.apostolidis@wur.nl; michail.tsiledakis@wur.nl;

1 Introduction

Pseudomonas aeruginosa PA14 is a gram-negative bacterium that is able to adapt to a range of environmental conditions, from aquatic environments to biofilms and host tissues. For this study, we analyzed an RNA-seq dataset of P. aeruginosa PA14, grown under 14 different conditions.Initially, the raw expression data were normalized using DESeq2, which provided a strong foundation for comparative transcriptomic analysis.Given the diversity of environmental stresses represented in the dataset, we formulated a set of null hypotheses to guide the analysis.

- $\mathbf{H}_0^{(1)}$**:** Genes associated with related stress-response processes do not exhibit coordinated expression patterns across experimental conditions.
- $\mathbf{H}_0^{(2)}$**:** Correlation-based network inference does not reveal modular organization corresponding to distinct biological functions.
- $\mathbf{H}_0^{(3)}$**:** The inferred gene co-expression network does not display non-random topological properties and is indistinguishable from a random network with similar size and density.

Rejecting these hypotheses would confirm that the network is not random and actually groups genes by their biological functions.

2 Materials & Methods

The dataset consists of a gene expression matrix comprising 176 genes and 47 samples. Initial quality control included verification of matrix dimensions and screening for non-finite values (NA/NaN/Inf). Expression values were $\log_2(x+1)$ transformed to stabilize variance and reduce the influence of highly expressed genes.

Missing values were handled by gene-wise mean imputation on the log-transformed data. Genes with more than 20% missing values or zero variance across samples were excluded to prevent undefined correlations; however, no genes met these criteria. Sample-level consistency and normalization were assessed with standard exploratory visualizations, including distribution plots, correlation heatmaps, and hierarchical clustering.

A gene co-expression network was inferred using Pearson correlation between gene expression profiles across all 47 samples. Correlation-based inference captures coordinated gene regulation across different experimental conditions without assuming prior regulatory structure(with the potential of missing non-linear interactions).

To obtain an interpretable network, the fully connected correlation matrix was sparsified using a data-driven approach. Three candidate thresholds corresponding to the top 2%, 5%, and 10% absolute correlation values were evaluated (Table 1). The top 5% threshold was selected as it balances network fragmentation and excessive density, yielding a strong global backbone network for downstream topological and condition-specific analyses.

Table 1 Network statistics for three candidate correlation thresholds.

Top proportion	Implied $\lvert r \rvert$	Nodes	Edges	Components	Largest component	Density
0.02 (top 2%)	0.8866	109	308	15	24	0.0523
0.05 (top 5%)	0.7978	144	770	8	56	0.0748
0.10 (top 10%)	0.6828	167	1540	1	167	0.1111

3 Results & Discussion

For hypothesis evaluation, we first examined global expression patterns across samples using dimensionality reduction and correlation analysis. Principal component analysis (PCA) was performed to explore overall structure and verify that biological replicates clustered together before building the network and showed that the first two principal components (PC1 and PC2) together explained approximately 54% of the total variance in gene expression. Samples generally separated according to stress condition, although some overlap between conditions was observed. The sample–sample correlation matrix demonstrated strong agreement among biological replicates. In particular, anoxic replicates showed high pairwise similarity, with Pearson correlation coefficients of approximately $r \approx 0.94$, supporting the presence of coherent condition-specific expression signatures.'

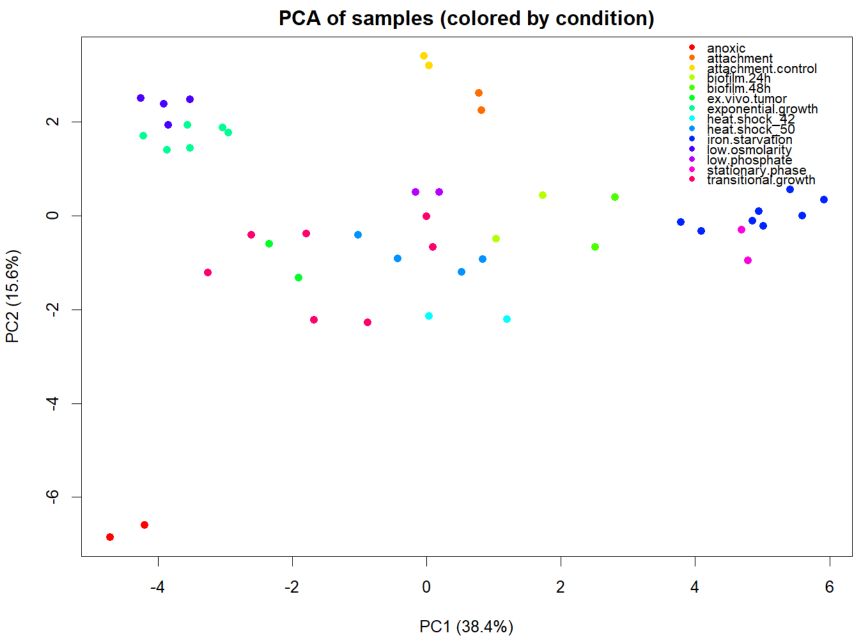

Fig. 1 PCA of samples based on $\log_2(x+1)$-transformed, imputed expression data. Points are colored by experimental condition, showing partial separation of stress conditions and tight grouping of most biological replicates.

The separation of samples by condition is visualized in Figure 1, where most biological replicates cluster closely, supporting coherent condition-specific expression signatures.

To highlight the strongest condition-responsive signals, we visualized the top 50 most variable genes (row-scaled) across samples (Figure 2), revealing blocks of coordinated up- and down-regulation across subsets of conditions.

The inferred co-expression network topology consisted of 144 nodes (genes) and 770 edges and organized into 8 components , with the largest containing 56 genes. *Biologically*, this fragmentation is a direct outcome of our threshold choice, meaning that if we were to choose the top 10% level, we would have a network which would merge components into a single "hairball"(1540 edges), while choosing a stricter threshold like the top 2%, could produce more than 8 components (specifically 15).

A Cytoscape visualization of the inferred global backbone network is shown in Figure 3, illustrating the modular structure and the fragmentation induced by the chosen sparsification threshold.

Hubs in the network were identified, with the highest connectivity included PA14_08300(with degree 25), PA14_08070 (with degree 24) and PA14_08280 (with degree 24), which suggests that these genes play central roles across multiple functional clusyers. The network displayed a clustering coefficient of 0.655, indicating dense

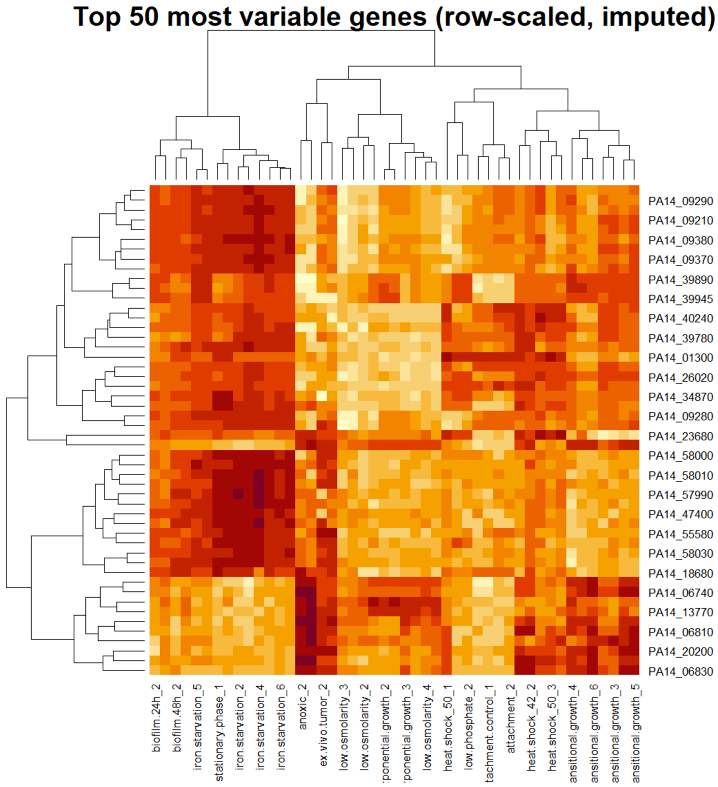

Fig. 2 Heatmap of the 50 most variable genes across the 47 samples (row-scaled) after $\log_2(x+1)$ transformation and imputation. Hierarchical clustering of genes and samples reveals groups of co-varying genes and condition-associated expression patterns.

neighborhoods which are true for biological pathways rather than random noise. To assess whether the inferred network shows scale-free-like organization, we examined the degree distribution on log–log axes (Figure 4). The weak linear fit ($R^2 = 0.199$) indicates that a power-law provides only a limited approximation in this network, something expected given the small network size and the correlation-threshold-driven construction.

The clusters were identified by visual inspection and BiNGO enrichment. The identification of distinct functional modules supports the hypothesis that coordinated gene expression underlies specific stress-response programs. Modules associated with iron acquisition, energy metabolism, and biofilm formation corresponded to biologically relevant conditions, indicating that the inferred network captures meaningful functional organization rather than random correlations.

Three clusters showed interesting biological relevance. The Iron Acquisition Cluster showed the strongest enrichment ($p \approx 1.9\times10^{-9}$), directly matching the iron starvation condition in which P. aeruginosa PA14 was subjected. The cluster represents genes that are responsible for producing Pyochelin, a siderophore used to scavenge iron and transporting it back to the cell. This is a critical mechanism for the bacteria when

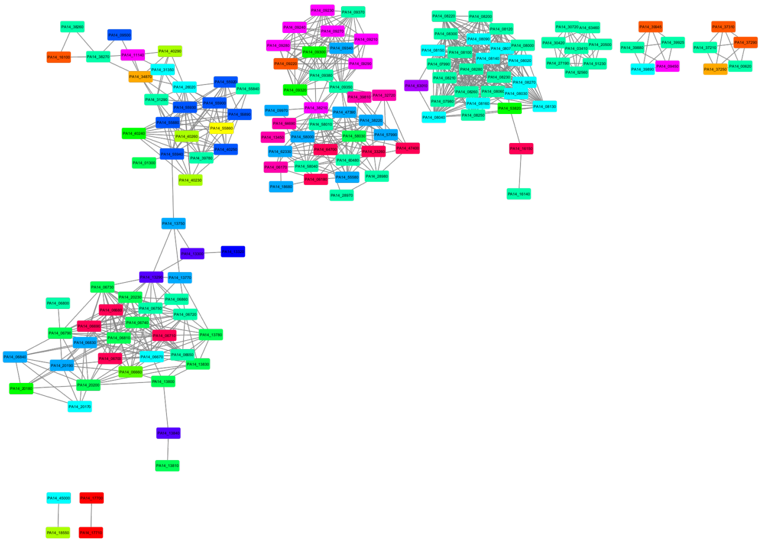

Fig. 3 Cytoscape visualization of the global gene co-expression backbone network (top 5% absolute Pearson correlations). Nodes represent genes and edges represent strong co-expression relationships. The layout highlights modular organization and multiple components under the selected threshold.

trying to survive in low-iron environments. Furthermore, another interesting cluster is Energy Metabolism ($p \approx 1.5 \times 10^{-6}$) , showing enriched genes that are involved in the ETC (electron transport chain) and anaerobic respiration(nitrate respiration). This shows the metabolic flexibility of bacteria when exposed to low oxygen conditions, e.g., anoxic condition, switching to nitrate respiration to survive, but also supporting the biofilm formation condition, as mature biofilms have low oxygen zones, making these genes upregulated to maintain metabolic capability while finally satisfying the stationary phase condition because as cultures grow dense, oxygen runs out triggering the same "low oxygen" metabolic shift. Lastly, cluster 3 is also valuable(corrected $p < 1.7 \times 10^{-2}$) , as it is unusual. It contains Pf4 prophage genes that are often activated during stress or biofilm formation. The enriched status of genes likely means that the Pf4 prophage was activated. The other clusters could also be interesting but due to time they were not analyzed further other than being annotated.

Together, these findings allow rejection of $H_0^{(1)}$ and $H_0^{(2)}$, while $H_0^{(3)}$ is only partially supported due to limitations in network size and thresholding. To validate the transcriptomic findings it would be best to also analyse proteomics data to confirm high correlations- particularly in the Iron Acquisition and Energy Metabolism clusters- as mRNA levels do not always correspond with protein abundance. Lastly, wet-lab experiments would validate the computational results by for example, the deletion of the hub gene PA14_08300 and measuring its growth under iron-limited conditions can solidify

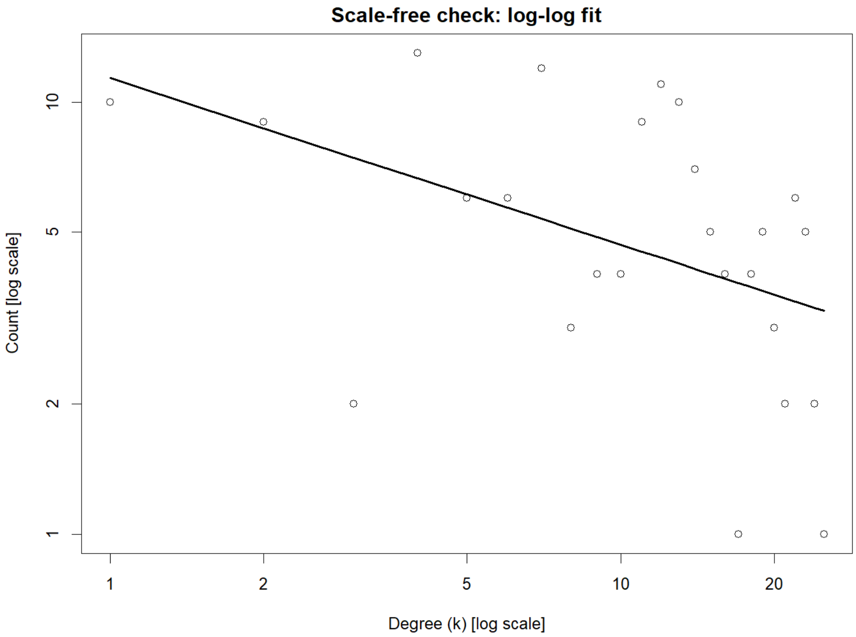

Fig. 4 Scale-free evaluation of the inferred network. The log–log degree distribution is shown together with a linear fit. The limited goodness-of-fit suggests that the degree distribution is not strongly consistent with a pure power-law in this thresholded, relatively small network.

findings in its central role in the iron acquisition network. Analogous procedures can be performed for every cluster.

Use of Generative AI

Generative artificial intelligence tools (GPT-5.2, and Gemini) were used only for approved purposes during the development of this assignment, such as text refinement (spelling, grammar, and clarity checks) and limited code assistance for troubleshooting and general R syntax guidance. The final report or analysis did not explicitly contain any AI-generated text, data interpretation, figures, results, or analytical conclusions. The authors independently created, examined, and verified all biological interpretations, analytical choices, scripts, figures, and textual content. AI tools were not employed for reference generation or citation selection, and the usage of generative AI was restricted, transparent, and recorded in compliance with the course rules. The final content, methodology, and conclusions of this study are entirely the authors' responsibility.

Table 2 Summary of functional modules identified in the gene co-expression network.

Cluster	Approx. size	Enriched biological processes	Associated conditions
Iron acquisition	∼30 genes	Pyochelin biosynthesis, iron transport	Iron starvation
Energy metabolism	∼25 genes	Ubiquinone biosynthesis, nitrate metabolism	Anoxic, biofilm, stationary
Glyoxylate metabolism	4 genes	Glyoxylate and glyoxylate catabolism	Stationary, biofilm
Prophage / stress response	∼15 genes	Viral reproduction, host interaction	Biofilm
Nitrogen scavenging	5 genes	Urea / allophanate metabolism	Iron starvation
Other minor clusters	$<$ 10 genes	Mixed or unannotated functions	Various